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CHAPTER 1: INTRODUCTION

1.1 Abstract

Online Car Marketplace website is basically made for providing the used vehicles to the customer which had one or more than one owner i.e. preowned or second-hand used car. This used car website is a huge platform on which seller can add his vehicle with complete information for selling and the buyer can buy the car according to their needs and choice. This website is basically aimed to provide vehicle with a complete information of the car i.e. brand, owner, location, description, price, etc.

1.2 Existing System and Need for System

2.1Existing System

- People buy and sell used cars through dealers, newspaper ads, or social media, which lack transparency, fair pricing, and proper comparison options..
- .

2.2Need for System

An online platform is needed to provide a trusted space where sellers can list cars with full details, and buyers can easily search, compare, negotiate, and book cars securely.

Scope of System

❖ Scopes

- Provides a platform for buying and selling used cars.
- Allows sellers to list their cars with full details (brand, price, location, etc.).
- Buyers can search, compare, and make offers on cars.
- Admin can manage users, sellers, and listings.

Admin Profile:

- Admin can register into system.
- Admin can log in into system.
- Admin can view details of the car.
- Admin can view the all customers details.
- Admin can view sold and buy car details.
- Admin can log out.

Buyer

- Buyer can buy the car by his own choice
- Buyer can send his affordable price request to the seller
- Buyer can search the car .
- Buyer can modify own profile.
- Buyer can log out.

Seller:

- Seller can add his car for selling.
- He can view his own added cars
- He can update and delete the car
- He can also view his own sold cars list with his buyer/customer details.
- He can response to the offer price request which is send by the buyer on his car
- Customer can log out.

Car Listing & Search Module:

- Display all available cars.
- Advanced filters (brand, price, owner type, etc.)

1.4 Operating Environment Hardware and Software:

- **Server-side requirement**

Software Requirement	**Hardware Requirement**
Operating System: Windows 7 or above	Processor: Intel Core i3 or above
Web Browser: Chrome, Mozilla Firefox	RAM: 2GB or above
Front End: Python	HDD: 512 GB or above
Back End: Java	
Database: MySql	

- **Client-side requirement**

Software Requirement	**Hardware Requirement**
Operating System: Windows 7 or above	Processor: Intel Core i3 or above
Web Browser: Chrome, Mozilla Firefox	RAM: 2GB or above
	HDD: 512 GB or above

1.5 Brief Description of Technology Used

The system leverages modern web development technologies to deliver a user-friendly and efficient platform for website generation. Below is a brief description of the technologies used:

- **Frontend Technologies**

  - **JavaServer Pages (JSP)**

    JavaServer Pages (JSP) is a server-side technology used to create dynamic, platform-independent web content. It allows developers to embed Java code directly into HTML pages, which the server processes to generate dynamic web pages based on user input or backend data..

- **Backend Technology**

  - **Java (Servlet)** :-

    Java Servlets are server-side programs that handle requests and responses in a web application. In the JobStream Online Job Portal, Java Servlets act as the **core backend technology** for processing business logic and managing communication between the **frontend** and the **database**.

- **Database**

  - **MySQL**:

    MySQL is an open-source **Relational Database Management System (RDBMS)** that uses **Structured Query Language (SQL)** to manage and manipulate databases. It's widely used in web applications and supports high performance, reliability, and ease of use..

CHAPTER 2: PROPOSED SYSTEM

2.1 Feasibility Study:

- **Technical Feasibility**

  The technology required to build the online car marketplace, like website development tools (HTML, CSS, Java, MySQL), is available. The project team should have basic knowledge of web development, databases, and hosting. Reliable internet and server systems ensure smooth functioning

- **Economic Feasibility**

  The project is affordable, with costs for development, hosting, and maintenance manageable. It can generate income through listing fees, ads, or premium features for sellers, making it a profitable investment in the long term.

- **Operational Feasibility**

  The platform is designed to be user-friendly, making it easy for buyers to search and sellers to list cars. It meets user needs efficiently, saving time and providing detailed car information, ensuring smooth operations for all users.

2.2 Objectives of the proposed system

- Objectives

  - To learn, implement and improve study of front-end and back-end languages.
  - To study normalization of database.
  - Implementing security measures to protect user data and prevent unauthorised access.

2.3 Users of the system

- **Admin**
- Register and log in to the system.
- View details of all cars.
- Monitor all customer and seller details.
- View sold and bought car transactions.
- Generate analytical reports.
- Log out securely.

.**Purpose:**
Ensures smooth platform operations, oversees user activities, and maintains data integrity.

- **Seller**

- Register and log in.
- Add vehicles for sale with complete details.
- View, update, or delete their listings.
- View sold car history along with buyer details.
- Respond to price offers submitted by buyers.
- Log out after session.

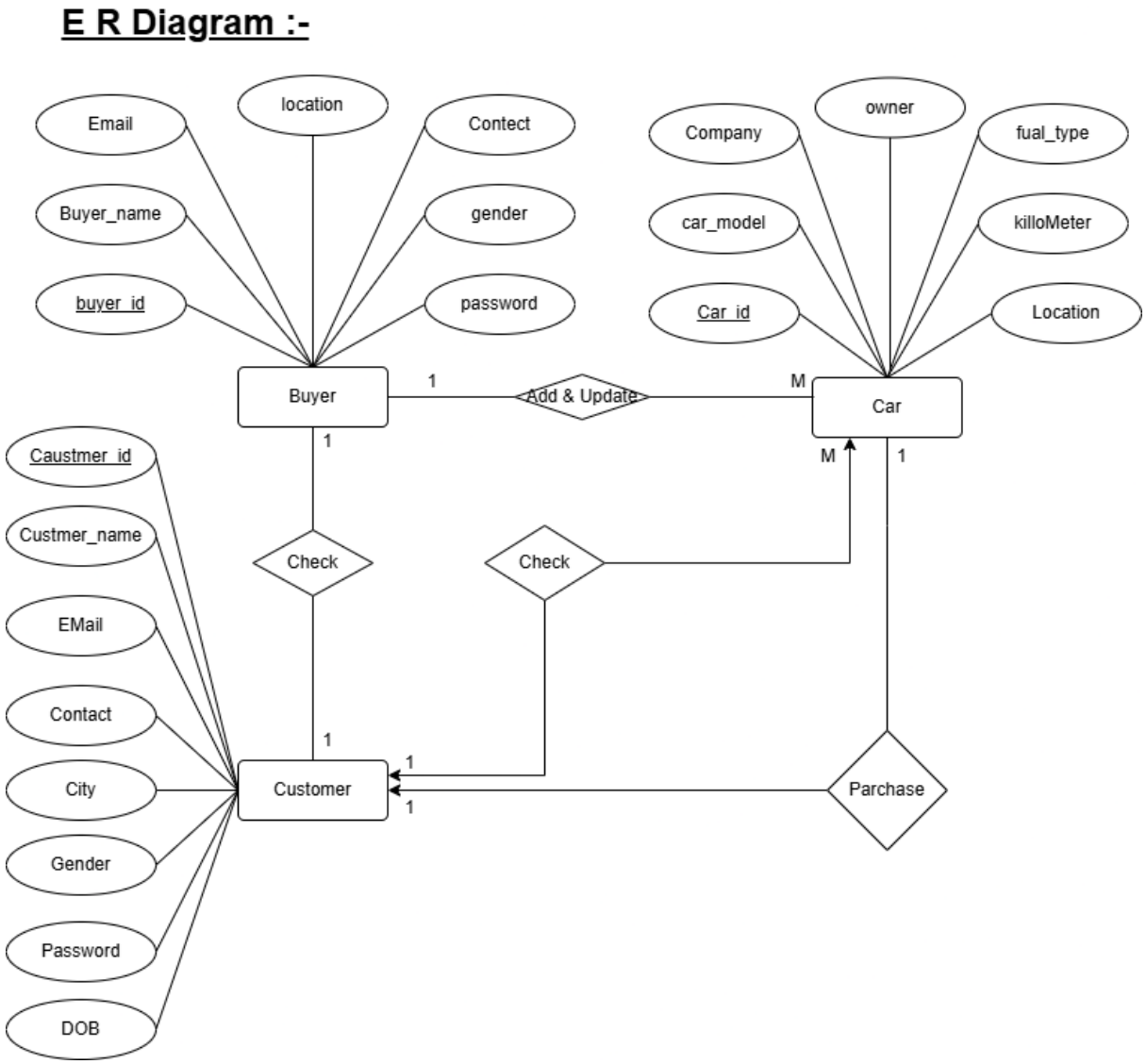
Purpose:
Enables individuals or businesses to list pre-owned cars and manage their sale process.

- **Buyer**
- Register and log in.
- Search cars using filters (brand, price, location, etc.).
- View detailed car listings.
- Send negotiable price offers to sellers.
- Modify personal profile.
- Securely book a car via the payment system.
- Log out after usage.

Purpose:
Facilitates buyers in exploring, comparing, and purchasing vehicles that meet their preferences.

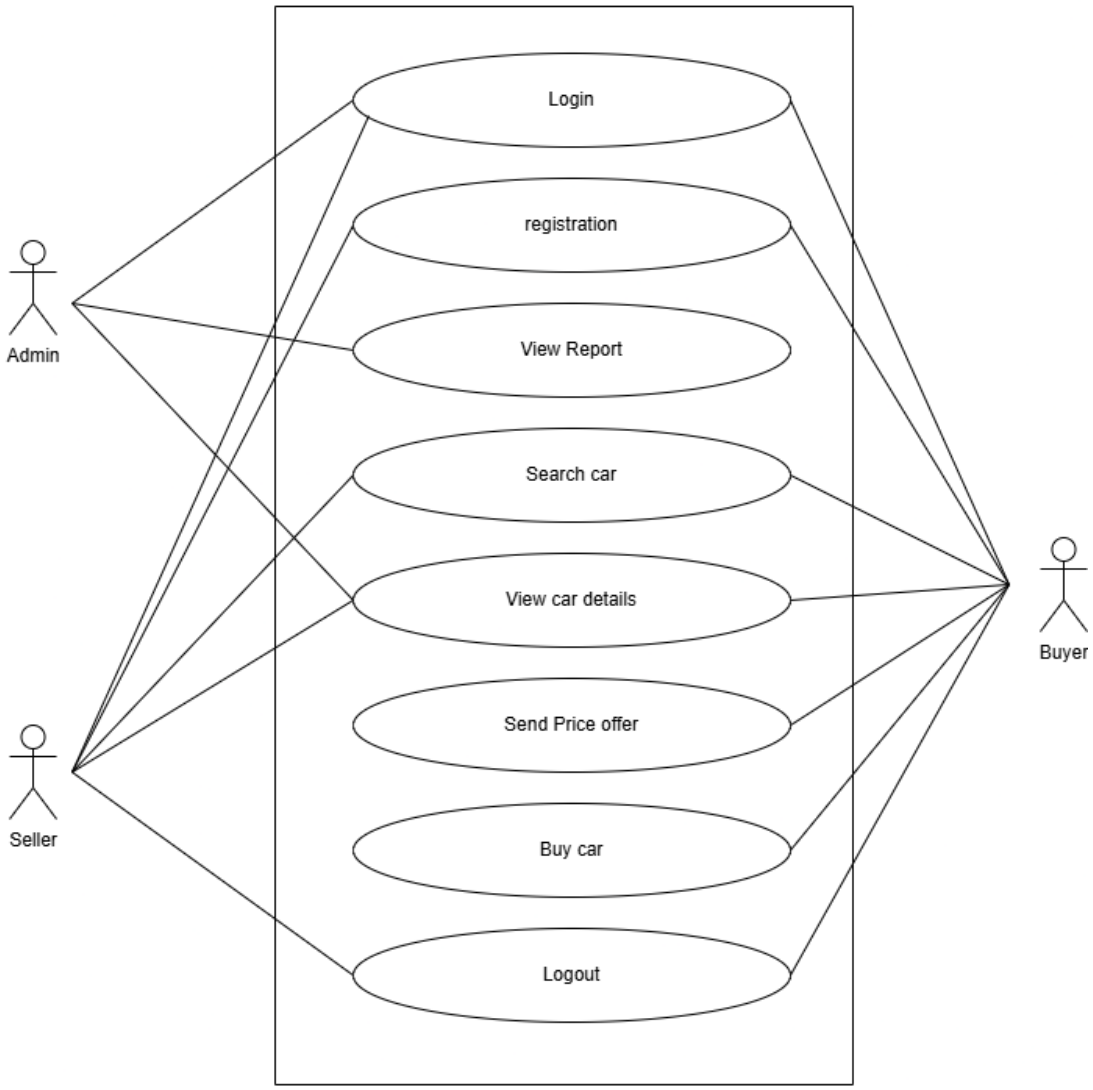
CHAPTER 3: ANALYSIS AND DESIGN

3.1 Entity Relationship Diagram (ERD)

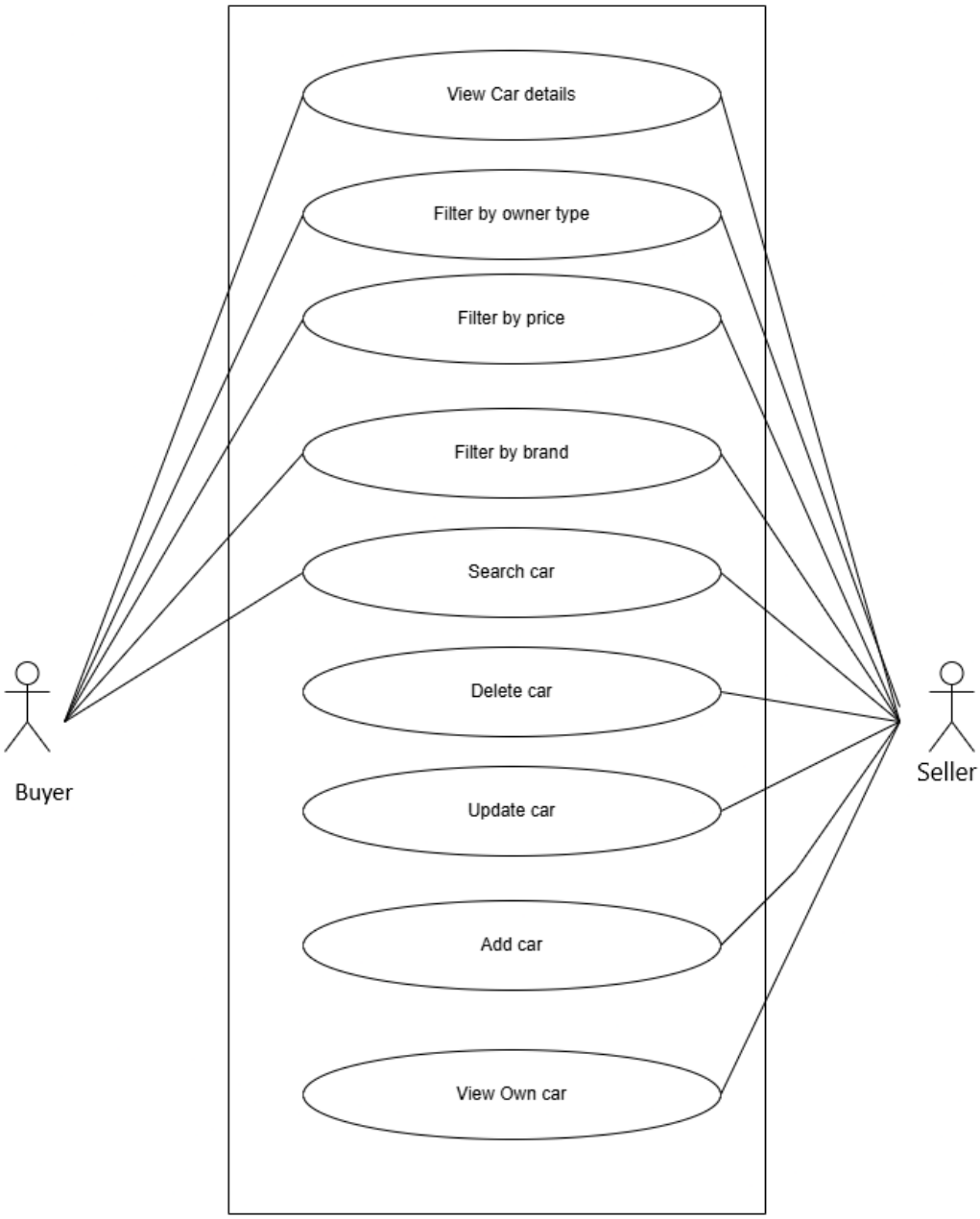


3.3 Use case diagrams

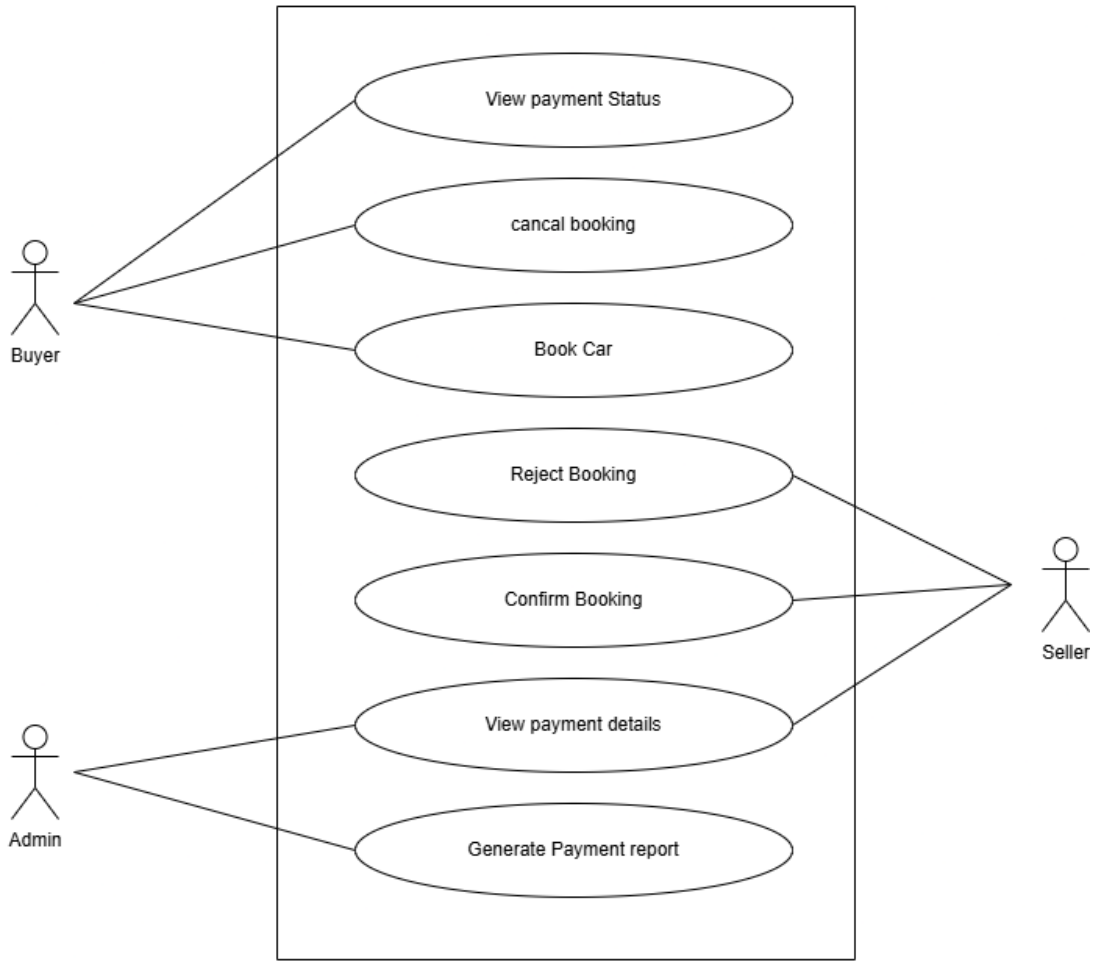
Use Case for System :-



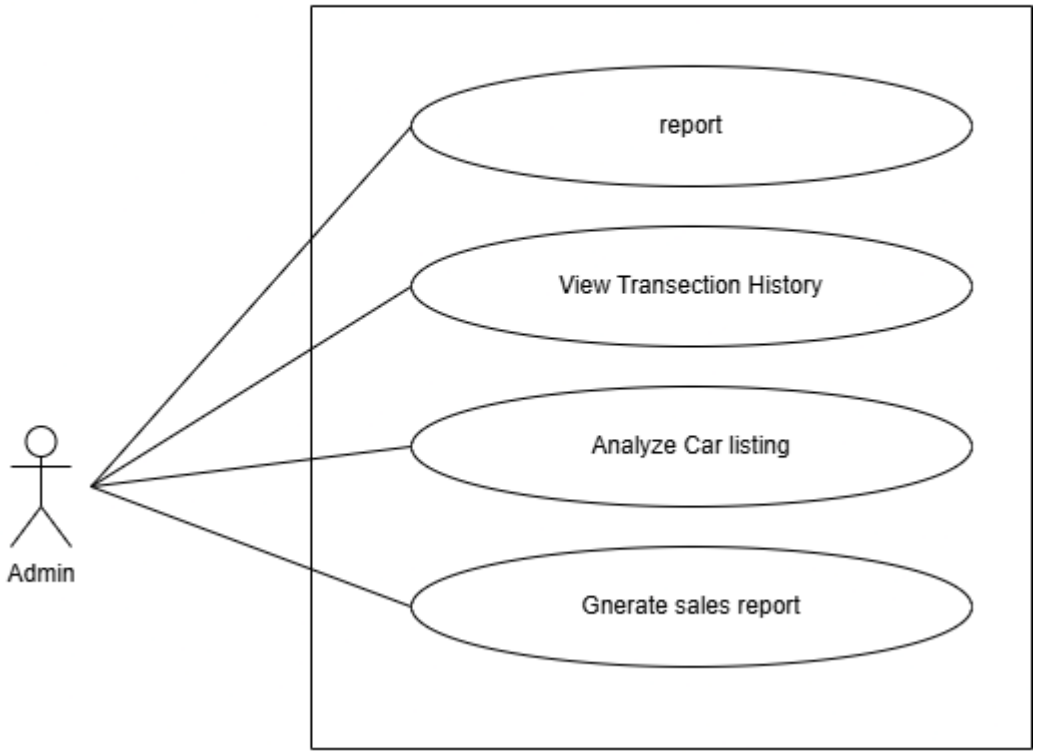
Use Case for Car listing and Search Module :-



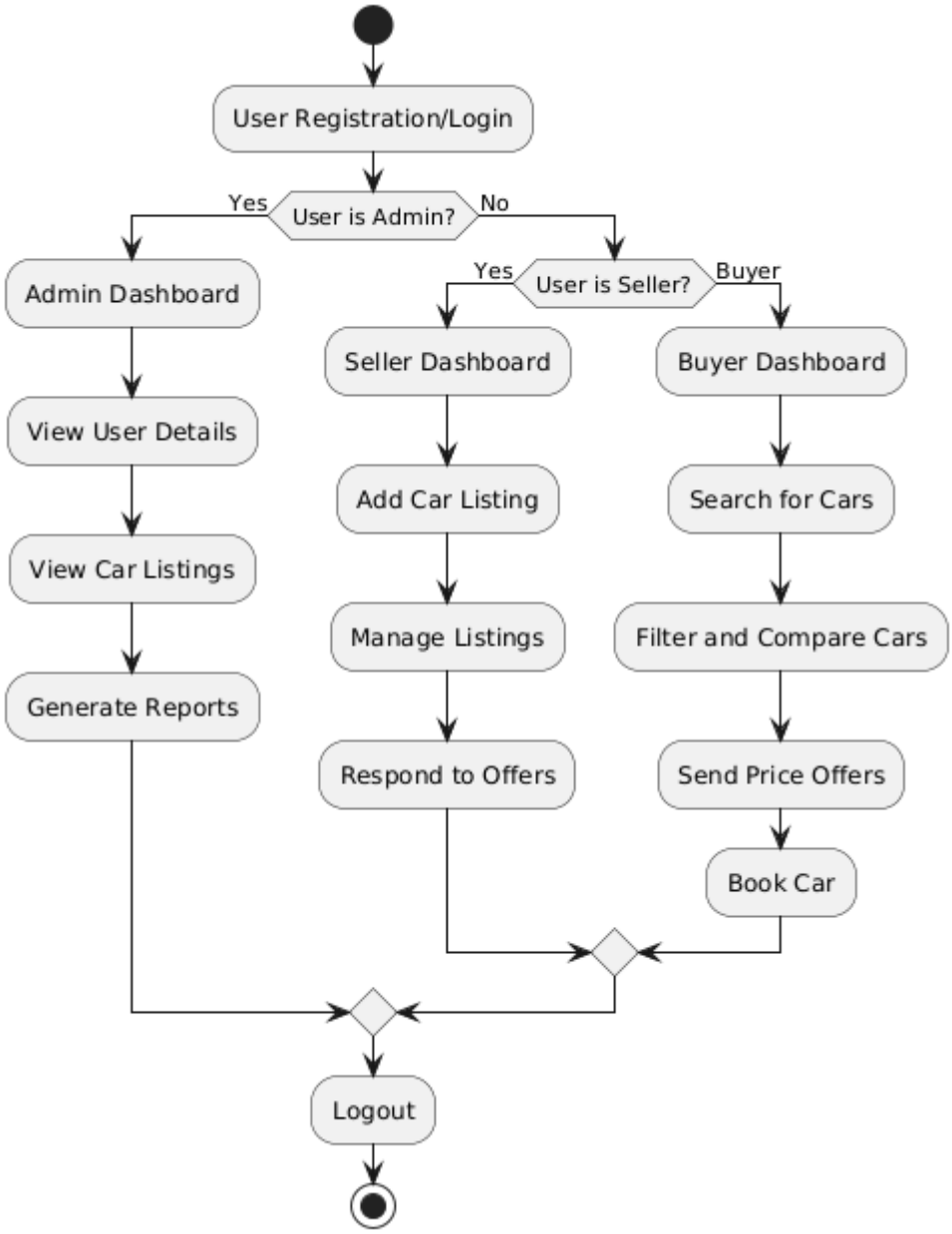
Use Case for booking:-



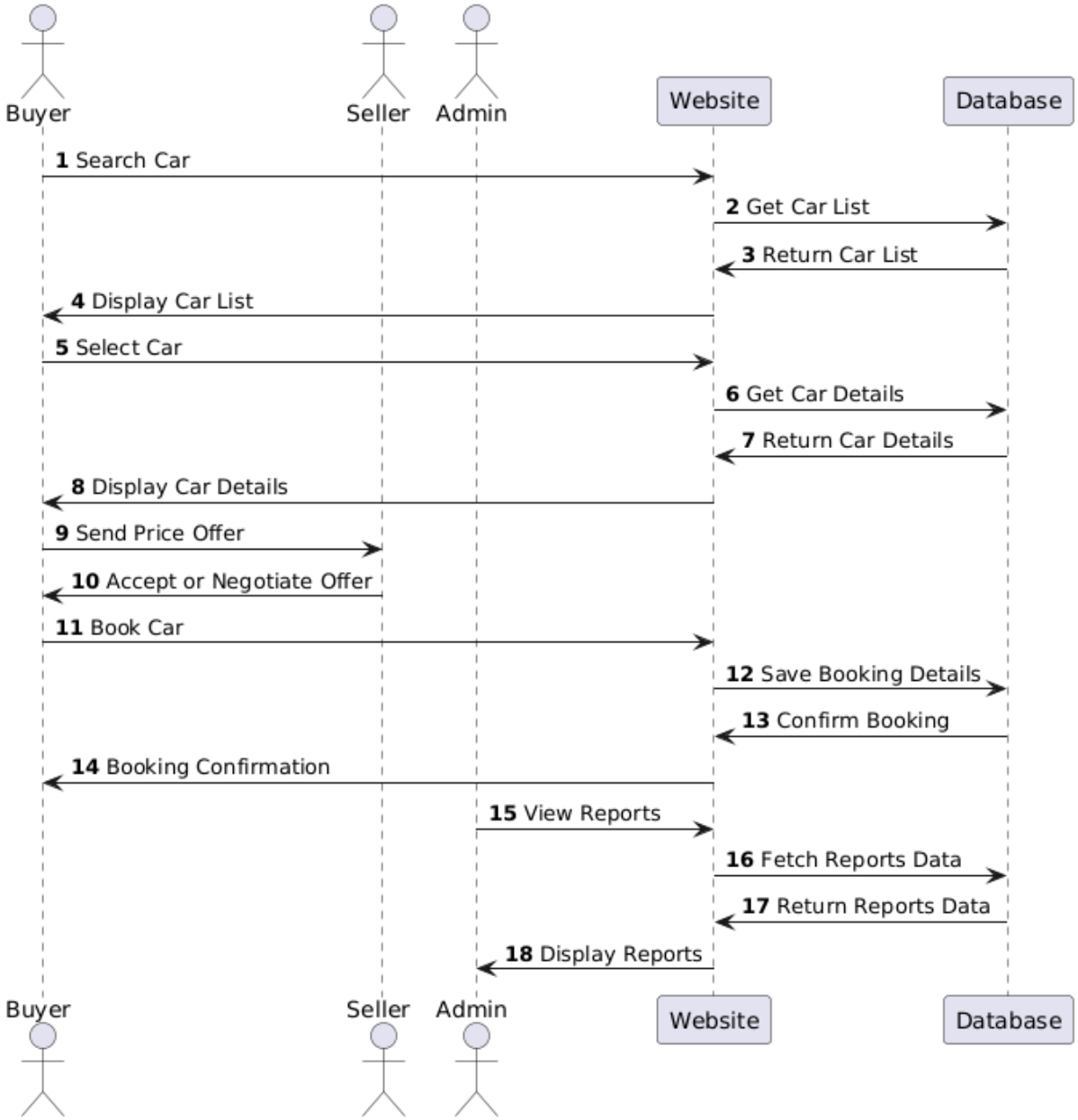
Use Case for report and analytic Module :-



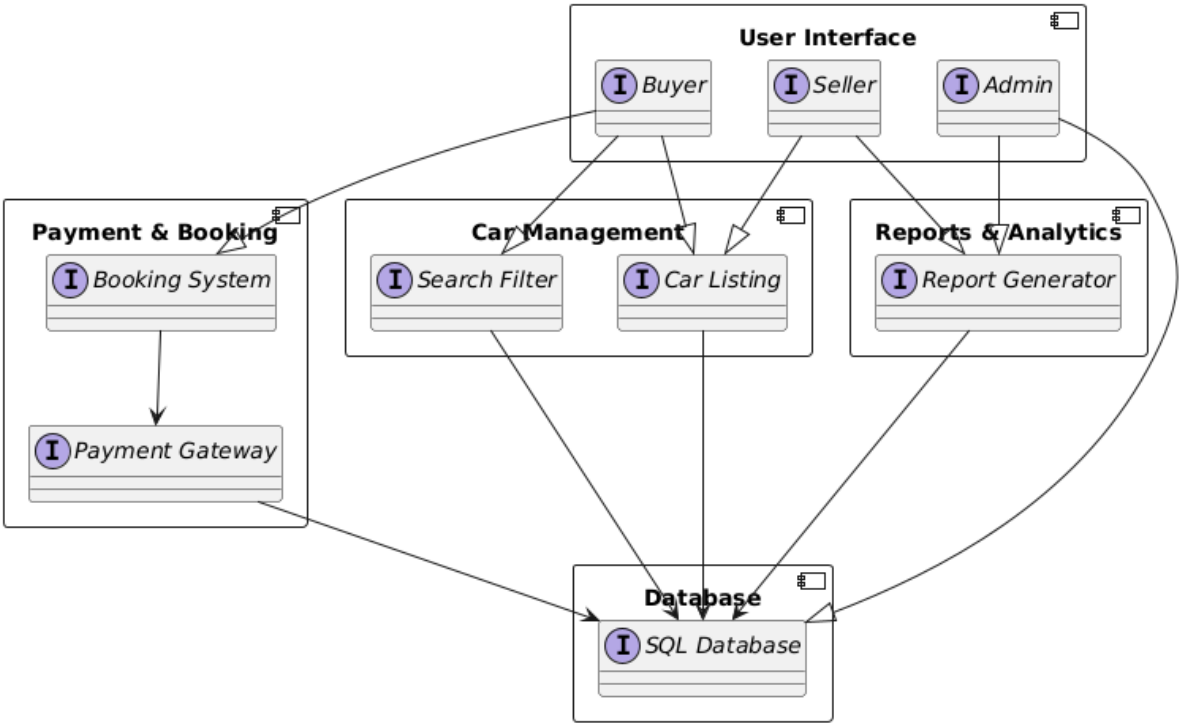
3.4 Activity diagram for system:



3.5 Sequence diagram for system:

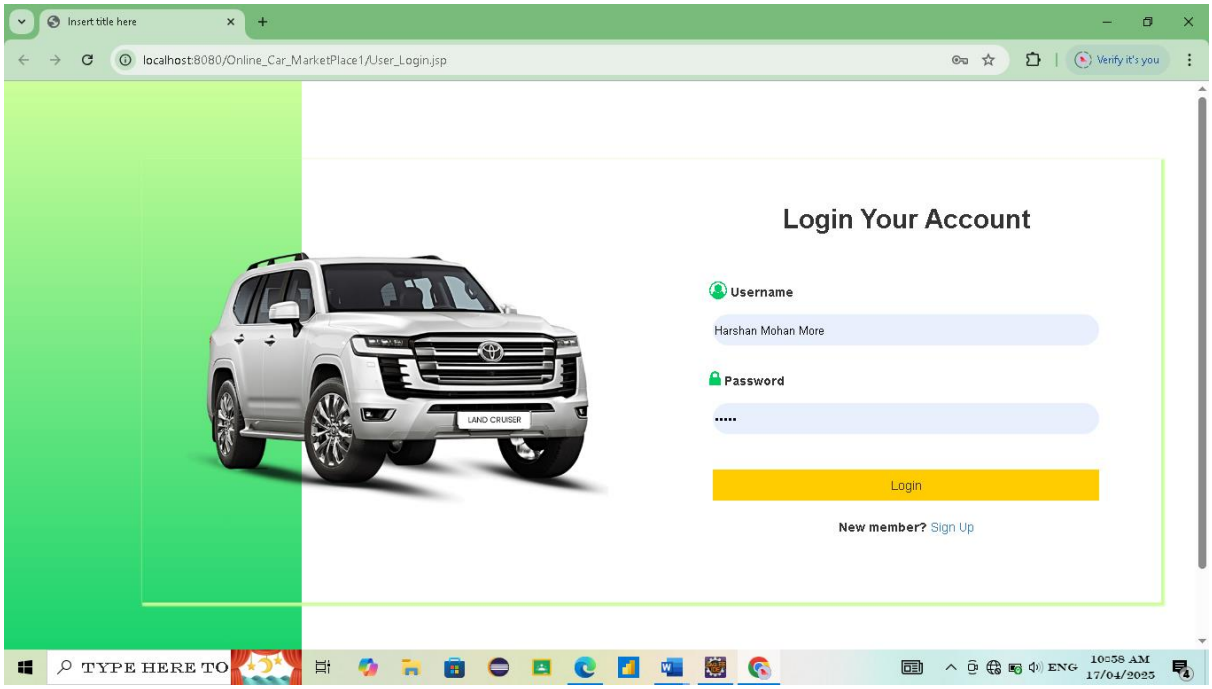


3.6 Component Diagram

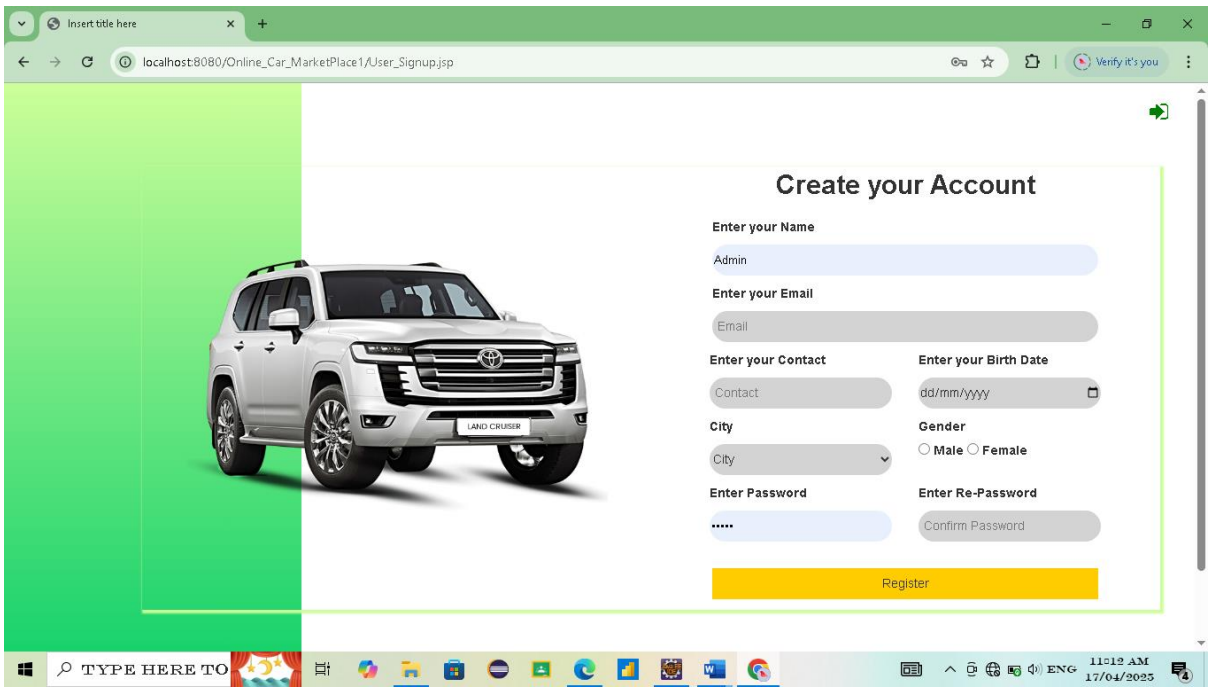


3.10 Sample Input and Output Screens

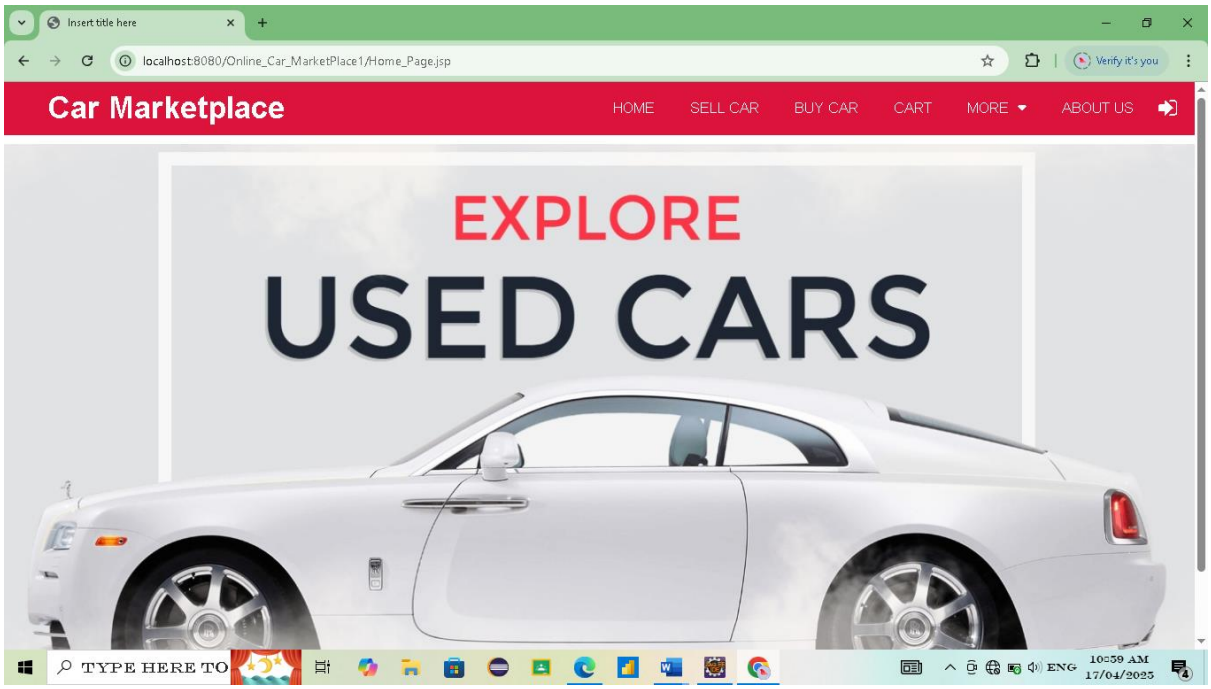
➢ User Login:



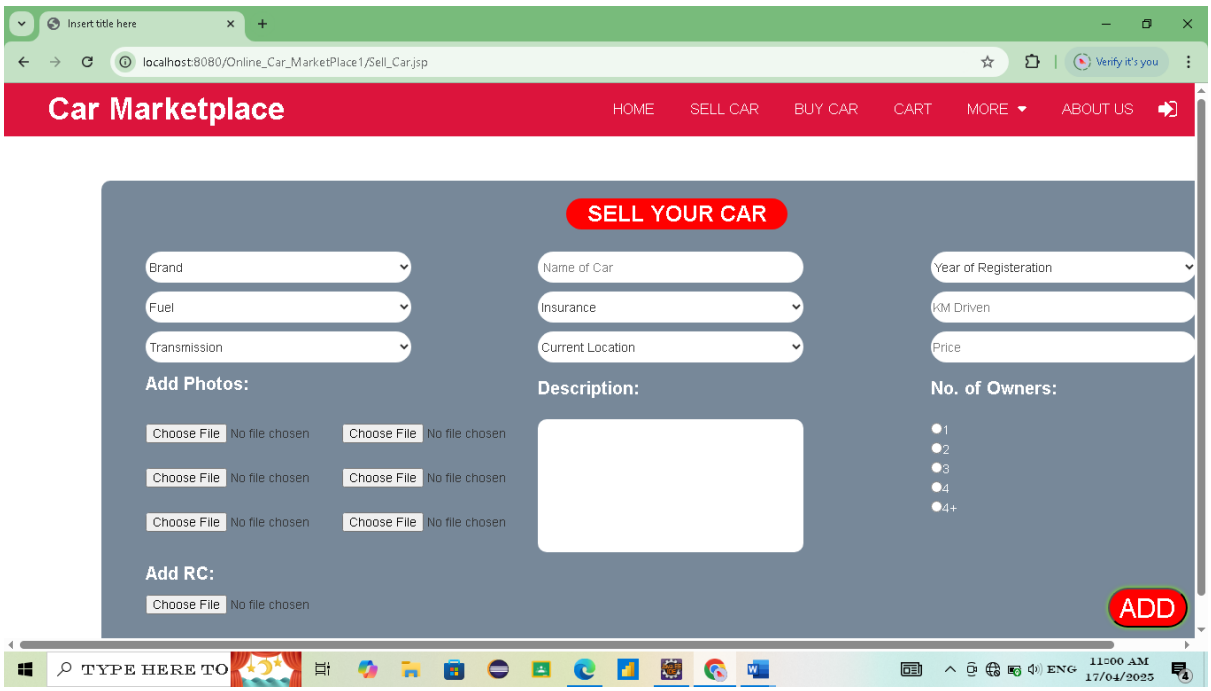
➢ Create Profile



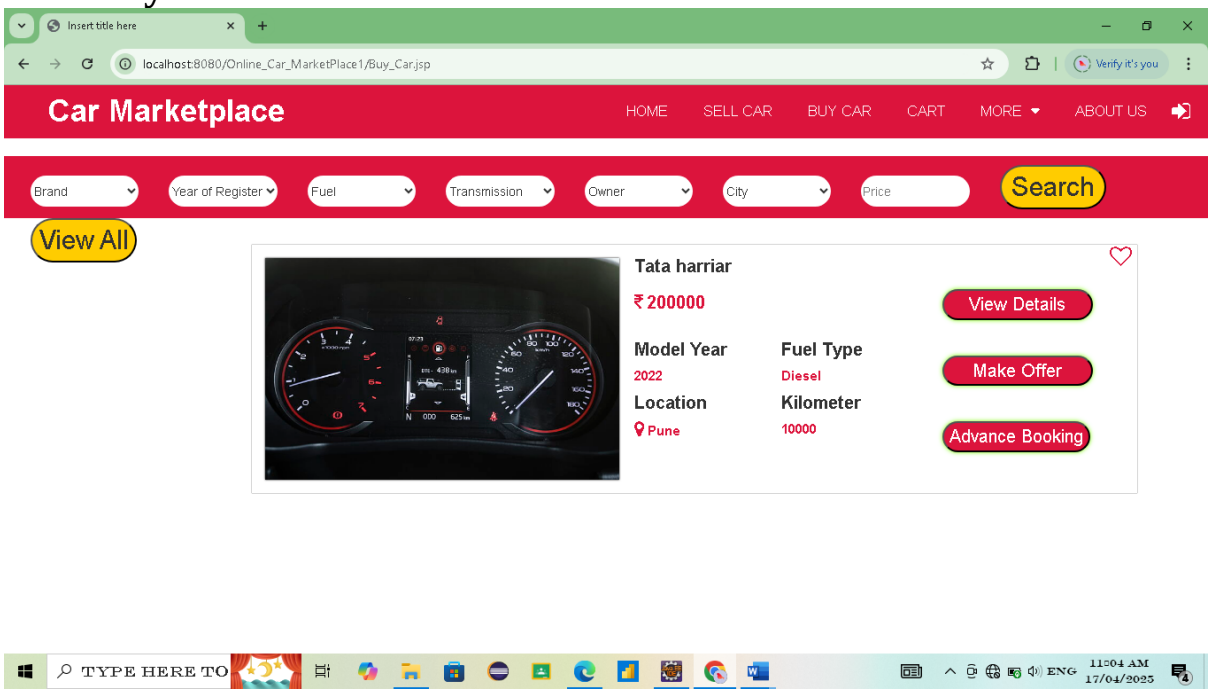
➢ User Homepage:



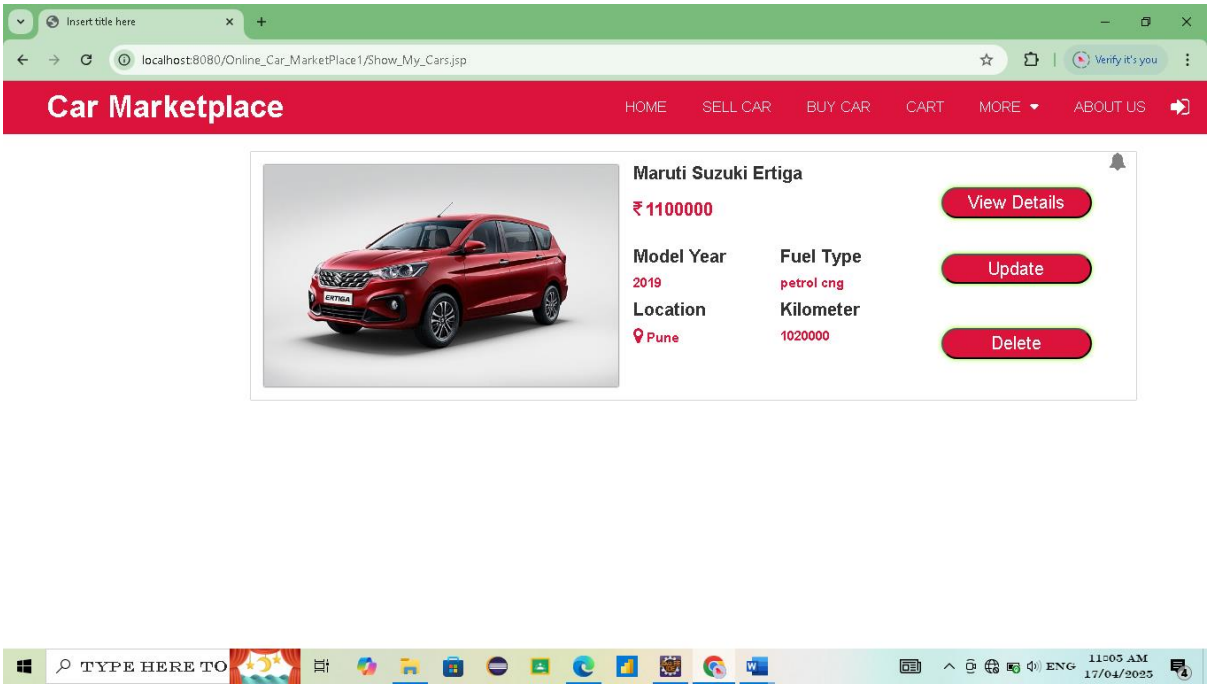
Sale car



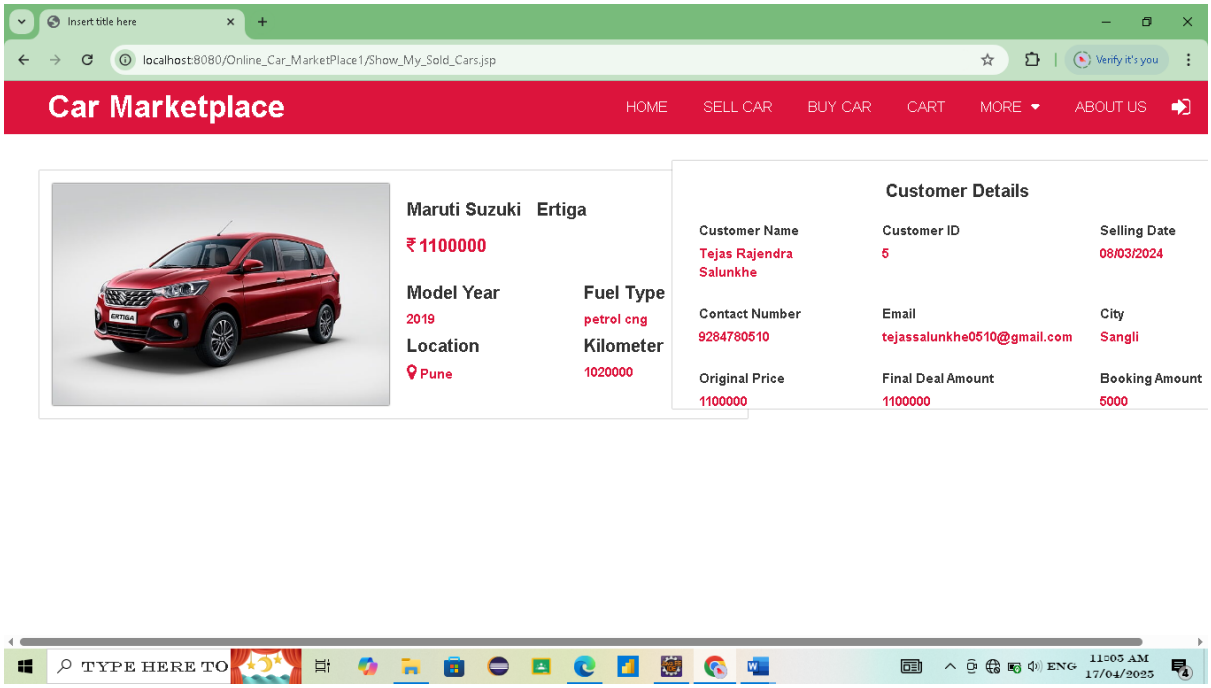
➢ Buy a Car:



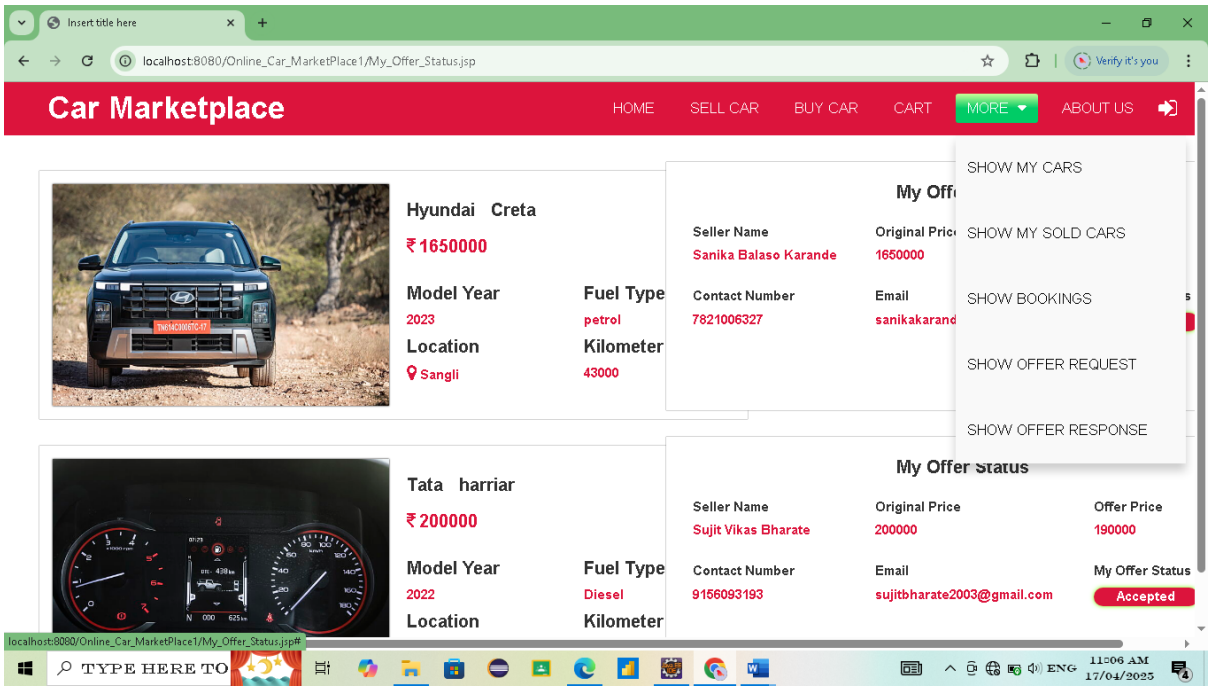
➢ My car page:



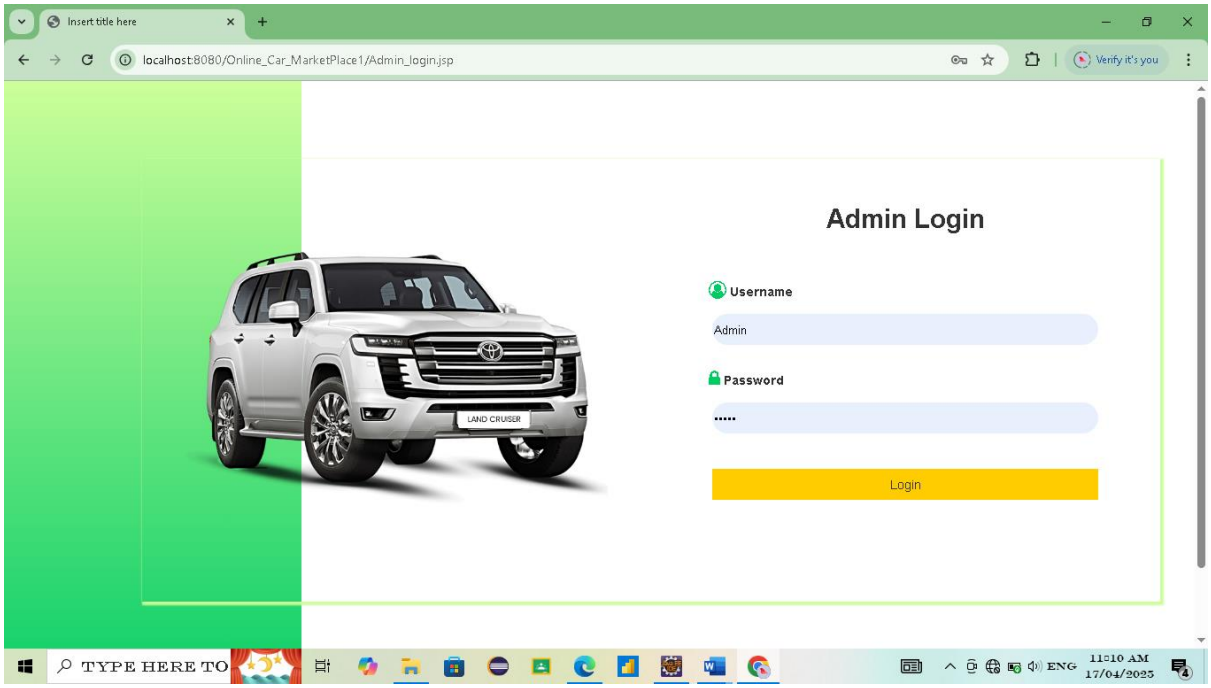
➢ Mysold Car :



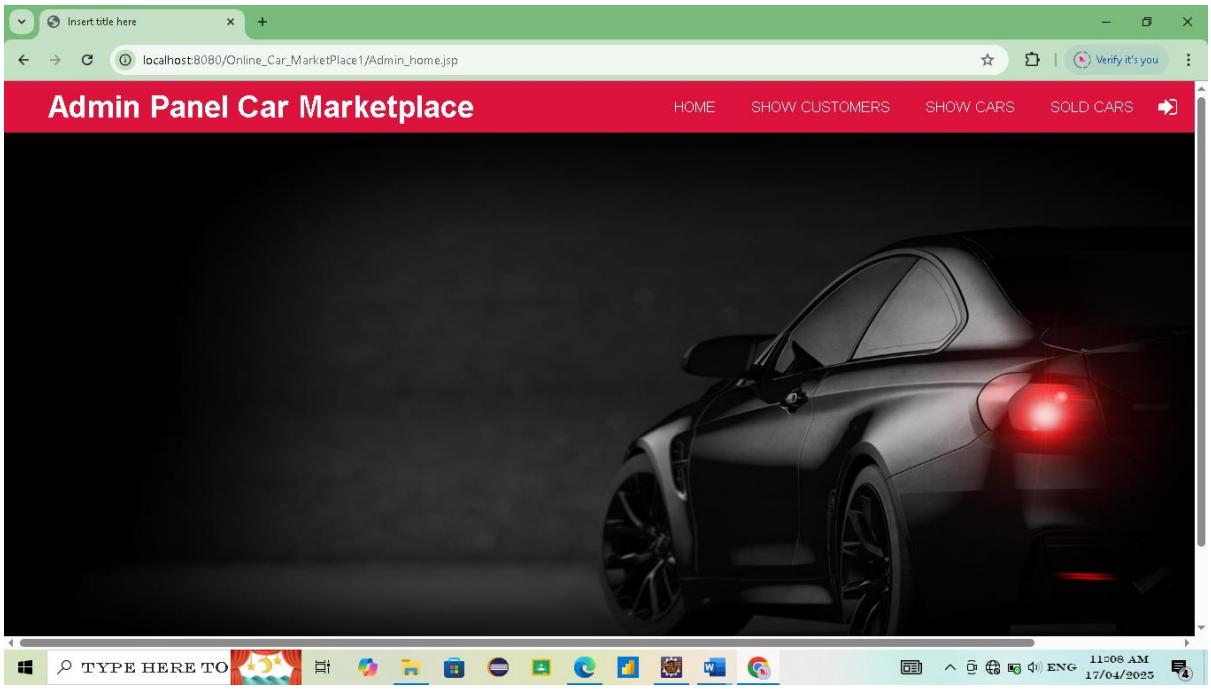
➢ Other options:



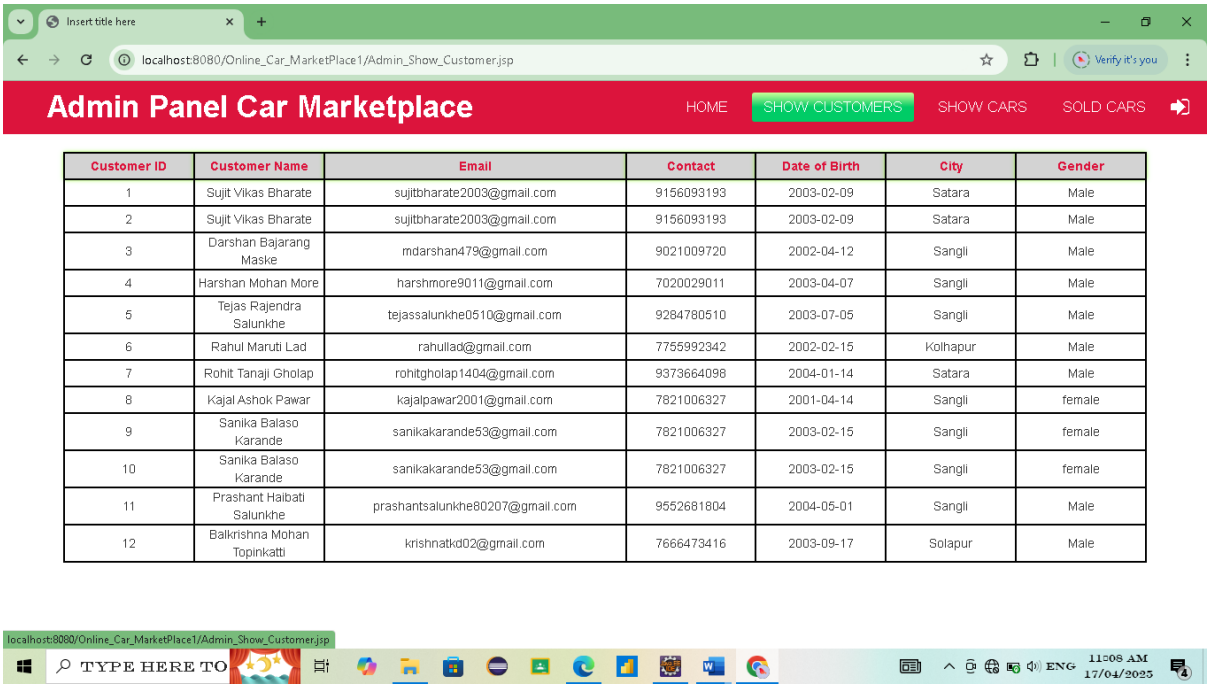
➢ Admin login



- **Admin homepage:**

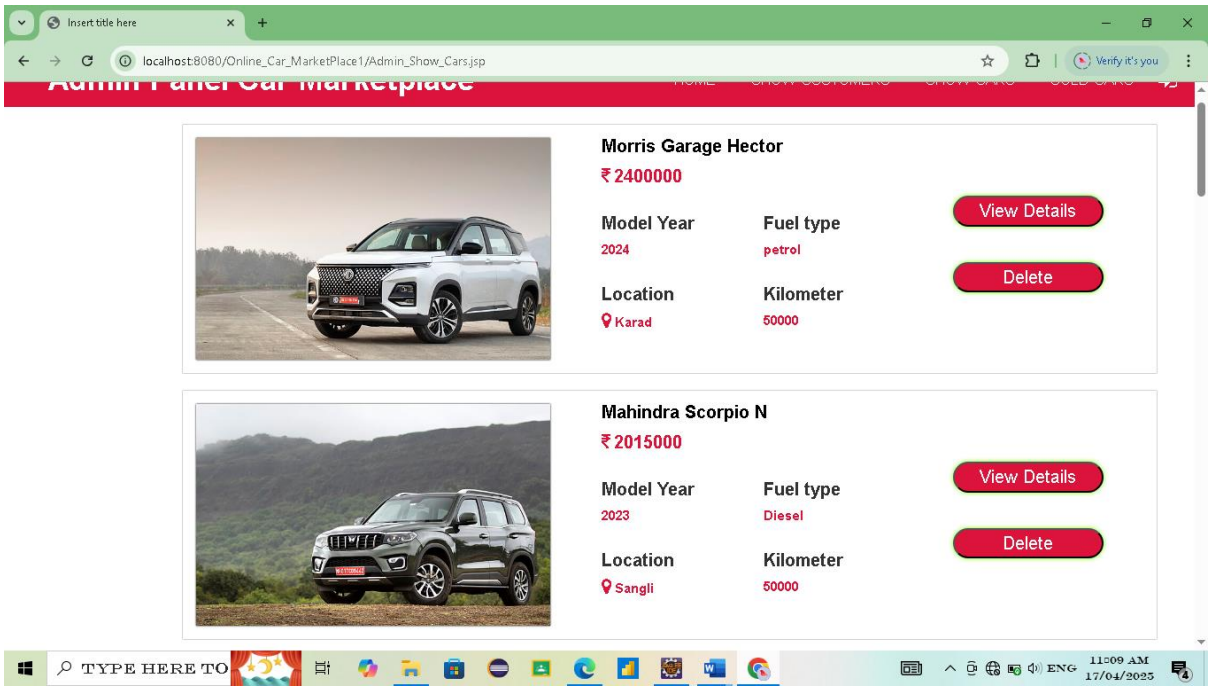


- **Show Customers :**

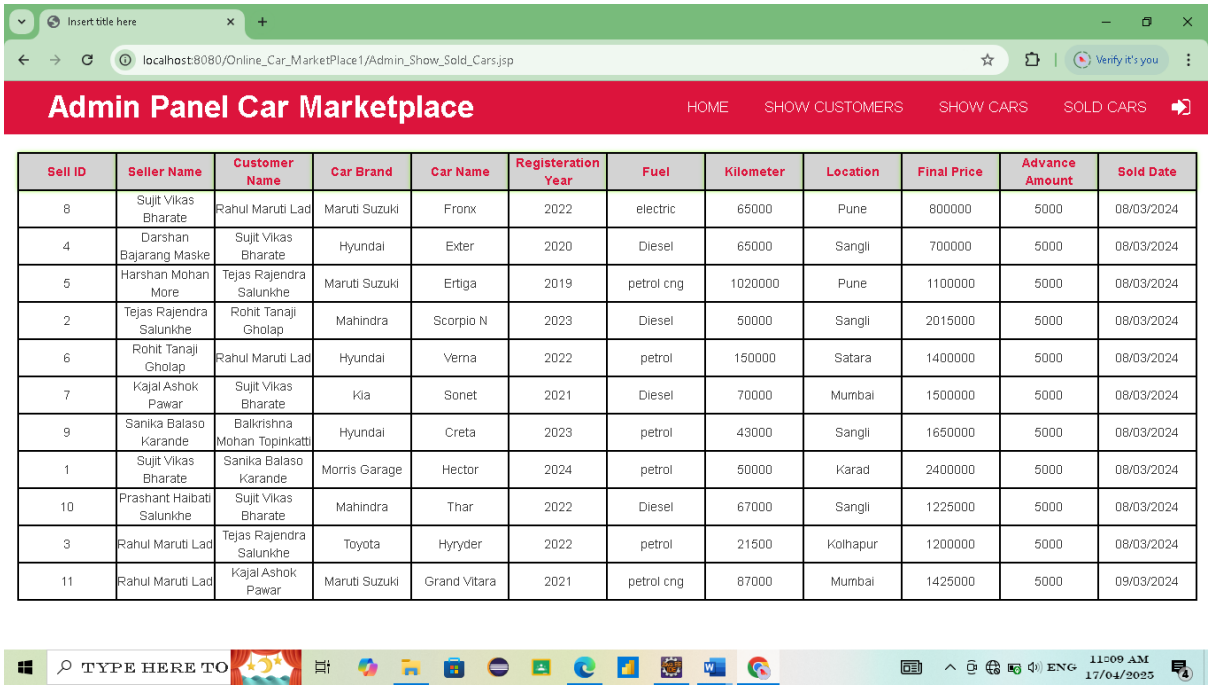


Customer ID	Customer Name	Email	Contact	Date of Birth	City	Gender
1	Sujit Vikas Bharate	sujitbharate2003@gmail.com	9156093193	2003-02-09	Satara	Male
2	Sujit Vikas Bharate	sujitbharate2003@gmail.com	9156093193	2003-02-09	Satara	Male
3	Darshan Bajarang Maske	mdarshan479@gmail.com	9021009720	2002-04-12	Sangli	Male
4	Harshan Mohan More	harshmore9011@gmail.com	7020029011	2003-04-07	Sangli	Male
5	Tejas Rajendra Salunkhe	tejassalunkhe0510@gmail.com	9284780510	2003-07-05	Sangli	Male
6	Rahul Maruti Lad	rahullad@gmail.com	7755992342	2002-02-15	Kolhapur	Male
7	Rohit Tanaji Gholap	rohitgholap1404@gmail.com	9373664098	2004-01-14	Satara	Male
8	Kajal Ashok Pawar	kajalpawar2001@gmail.com	7821006327	2001-04-14	Sangli	female
9	Sanika Balaso Karande	sanikakarande53@gmail.com	7821006327	2003-02-15	Sangli	female
10	Sanika Balaso Karande	sanikakarande53@gmail.com	7821006327	2003-02-15	Sangli	female
11	Prashant Haibati Salunkhe	prashantsalunkhe80207@gmail.com	9552681804	2004-05-01	Sangli	Male
12	Balkrishna Mohan Topinkatti	krishnatkd02@gmail.com	7666473416	2003-09-17	Solapur	Male

➢ Car list



Sold car records:



Sell ID	Seller Name	Customer Name	Car Brand	Car Name	Registeration Year	Fuel	Kilometer	Location	Final Price	Advance Amount	Sold Date
8	Sujit Vikas Bharate	Rahul Maruti Lad	Maruti Suzuki	Fronx	2022	electric	65000	Pune	800000	5000	08/03/2024
4	Darshan Bajarang Maske	Sujit Vikas Bharate	Hyundai	Exter	2020	Diesel	65000	Sangli	700000	5000	08/03/2024
5	Harshan Mohan More	Tejas Rajendra Salunkhe	Maruti Suzuki	Ertiga	2019	petrol cng	1020000	Pune	1100000	5000	08/03/2024
2	Tejas Rajendra Salunkhe	Rohit Tanaji Gholap	Mahindra	Scorpio N	2023	Diesel	50000	Sangli	2015000	5000	08/03/2024
6	Rohit Tanaji Gholap	Rahul Maruti Lad	Hyundai	Verna	2022	petrol	150000	Satara	1400000	5000	08/03/2024
7	Kajal Ashok Pawar	Sujit Vikas Bharate	Kia	Sonet	2021	Diesel	70000	Mumbai	1500000	5000	08/03/2024
9	Sanika Balaso Karande	Balkrishna Mohan Topinkatti	Hyundai	Creta	2023	petrol	43000	Sangli	1650000	5000	08/03/2024
1	Sujit Vikas Bharate	Sanika Balaso Karande	Morris Garage	Hector	2024	petrol	50000	Karad	2400000	5000	08/03/2024
10	Prashant Haibati Salunkhe	Sujit Vikas Bharate	Mahindra	Thar	2022	Diesel	67000	Sangli	1225000	5000	08/03/2024
3	Rahul Maruti Lad	Tejas Rajendra Salunkhe	Toyota	Hyryder	2022	petrol	21500	Kolhapur	1200000	5000	08/03/2024
11	Rahul Maruti Lad	Kajal Ashok Pawar	Maruti Suzuki	Grand Vitara	2021	petrol cng	87000	Mumbai	1425000	5000	09/03/2024

CHAPTER 4: CODING Sample code

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.Statement;

/**
 * Servlet implementation class User_Signup
 */
public class User_Signup extends HttpServlet
{
	Connection cn=null;
	Statement st=null;
	@Override
	protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException
	{
		// TODO Auto-generated method stub
			HttpSession session = req.getSession();
			Database db=new Database();
			String result=db.Connectdb();
			PrintWriter out=resp.getWriter();
			out.println(result);

			String username= req.getParameter("username");
			String email=req.getParameter("email");
			String contact=req.getParameter("contact");
			String dob=req.getParameter("dob");
			String city=req.getParameter("city");
```

```
                    String gender=req.getParameter("gender");
                    String password=req.getParameter("password");
                    String cpassword=req.getParameter("cpassword");
                    String event = req.getParameter("submit");

                    if(event.equals("Register"))
{
if (username.equals("") || email.equals("") || contact.equals("") || dob.equals("") ||
city.equals("") || gender.equals("") || password.equals("") || cpassword.equals("") )
{
resp.setContentType("text/html");
out.println("<script type=\"text/javascript\"> alert('Something is Empty..!!');
location='User_Signup.jsp'; </script>");
}
else
{
if(password.equals(cpassword))
{
String sql = "insert into user_signup
(customer_name,email,contact,dob,city,gender,password,cpassword) values
('"+username+"','"+email+"','"+contact+"','"+dob+"',
'"+city+"','"+gender+"','"+password+"','"+cpassword+"')";
String insert = db.Insert(sql);
out.println(insert);
resp.setContentType("text/html");
out.println("<script type=\"text/javascript\"> alert('Register Successfull');
location='User_Login.jsp';  </script> ");
}
else
{
resp.setContentType("text/html");
out.println("<script type=\"text/javascript\"> alert('Password and Confirm Password
are not matched..!!'); location='User_Signup.jsp'; </script>");
}
}
s
}



}
```

}

CHAPTER 5: LIMITATIONS OF SYSTEM

While the proposed system provides an efficient and user-friendly platform for generating business websites,

- **Lack of Physical Inspection:**
  - Buyers cannot physically inspect or test drive the vehicle through the platform.
  - Trust heavily depends on the seller's honesty and description accuracy.
- **Limited Verification of Listings:**
  - There may be challenges in verifying the authenticity of car details (like condition, ownership history, accident history) if not integrated with official databases**.**
- **Security Risks:**
  - Although there are security measures, risks like data breaches, hacking, and unauthorized access remain a concern, especially during payment transactions.
- **Dependency on Internet Connectivity:**
  - Users need a stable internet connection to access and use the platform effectively, limiting usability in remote areas.
- **Negotiation Limitations:**
  - Negotiations are limited to price offers through the system; there is no provision for real-time chats or in-depth discussions.

- **Fraudulent Activities:**

  - Possibility of fraudulent sellers or fake buyers despite basic verification mechanisms.

- **No After-Sale Support:**

  - The platform does not provide warranty, after-sales service, or return/refund policies, which are common in traditional dealership purchases.

- **Geographical Constraints:**

  - Listings and transactions may initially be limited to specific cities or regions, restricting nationwide accessibility**.**

- **Maintenance and Scalability:**

  - As the user base grows, maintaining server performance and ensuring fast loading times could become challenging without scalable infrastructure.

- **User Skill Requirement:**

  - Some users, especially less tech-savvy individuals, might find the platform's functionalities like filtering, payment, or negotiation confusing.

CHAPTER 6: PROPOSED ENHANCEMENTS

1. Vehicle Verification & Inspection Integration

- Enhancement: Integrate with RTO (Regional Transport Office) or government APIs to validate ownership, registration, insurance, and accident history.
- Benefit: Builds buyer trust and reduces fraud.

2. Real-Time Chat or Call Feature

- Enhancement: Implement secure chat or VoIP-based calling between buyers and sellers within the platform.
- Benefit: Facilitates better negotiation and immediate query resolution.

3. AI-Based Price Estimator

- Enhancement: Add a tool that suggests fair market value of a listed vehicle based on brand, model, year, condition, mileage, etc.
- Benefit: Prevents overpriced listings and helps buyers/sellers get a fair deal.

4. Vehicle Condition Checklist / Inspection Report

- Enhancement: Allow sellers to upload mechanic-verified condition reports or integrate with third-party vehicle inspection services.
- Benefit: Enhances transparency about vehicle quality.

5. Advanced Security Features

- Enhancement: Implement two-factor authentication (2FA), CAPTCHA, and encrypted payment gateways.
- Benefit: Secures user data and transaction information against threats.

6. In-App Notifications & Email Alerts

- Enhancement: Add real-time notifications for offers, price changes, or new listings that match user preferences.
- Benefit: Keeps users engaged and increases successful transactions.

7. Mobile App Version

- Enhancement: Develop a responsive Android/iOS mobile app for broader accessibility.
- Benefit: Reaches a larger user base and allows on-the-go usage.

8. Ratings and Reviews System

- Enhancement: Allow buyers to rate sellers and leave feedback.
- Benefit: Helps future users identify trustworthy sellers and increases platform credibility.

9. Logistics/Delivery Option

- Enhancement: Partner with third-party logistics for optional vehicle delivery.
- Benefit: Convenience for long-distance buyers and an added value proposition.

10. Premium Listings & Advertisements

- Enhancement: Offer featured listings for sellers to increase visibility or place ads.
- Benefit: New monetization stream for platform sustainability.

CHAPTER 7: CONCLUSION

The **Online Car Marketplace** project successfully delivers a robust, interactive, and scalable solution to bridge the gap between used car buyers and sellers. In today's digital age, the need for an efficient, trustworthy, and transparent platform for buying and selling pre-owned vehicles is critical. This system is designed to meet those needs by offering a comprehensive set of functionalities including car listing management, dynamic search filters, profile handling, and secure booking mechanisms.

Through its modular design and clear separation of user roles—Admin, Buyer, and Seller—the platform ensures that each user has tailored access and control over relevant features. Admins oversee the integrity and analytics of the platform; sellers can manage vehicle listings and negotiate offers; and buyers can conveniently browse, filter, and propose deals for vehicles that fit their preferences.

From a technical perspective, the project employs a solid full-stack foundation with **Java Spring Boot** for the backend, **HTML/CSS/JS with Bootstrap** for the frontend, and **SQL** for the database layer. This architecture allows for maintainability, reusability, and scalability. Features like user session management, secure login/logout, and profile updates align with modern web application standards.

Despite its strengths, the system has certain limitations, such as the absence of physical verification tools, real-time interaction, and integration with official vehicle databases. However, the proposed enhancements—such as AI-based price estimators, live chat modules, third-party inspection support, and mobile app development—offer promising pathways to overcome these shortcomings and elevate the platform’s usability and commercial potential.

Overall, this project not only fulfills the academic objectives of learning database normalization, implementing secure authentication, and understanding full-stack development, but it also mirrors the real-world needs of digital marketplaces. With strategic improvements and user feedback incorporation, the Online Car Marketplace could evolve into a viable commercial product offering a seamless and trusted car trading experience.

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